Research Overview

Yanghao Chen

My research focuses on computational mechanics and scientific machine learning. I have worked on language model fine-tuning for mechanics of materials, prediction of concentration and stress fields in battery materials, and physics-informed neural networks for shock capturing. My current research centers on hybrid finite element–neural operator solvers based on non-overlapping domain decomposition, including an ongoing extension to three-dimensional fiber-reinforced composites. I am also interested in GPU-accelerated scientific computing and differentiable programming for structural optimization and inverse problems.

LLM Fine-Tuning for Mechanics

My first research project at Tongji University, supervised by Professor Ying Zhao, aimed to improve a general-purpose language model's ability to answer questions and explain concepts in mechanics of materials. I constructed instruction datasets from textbooks and technical literature, covering stress analysis, constitutive relations, and failure theories. I organized each example into instruction, input, and output fields and adapted the data-loading code for training. I then fine-tuned Qwen2.5-7B using low-rank adaptation (LoRA). After training, I publicly released two instruction datasets, the fine-tuned model, and its merged version on Hugging Face.

Collaboration and Open Research

Following fine-tuning, I worked with my teammates to deploy the model locally for the China International College Students' Innovation Competition. To accommodate limited GPU memory, we reduced the model's numerical precision and adjusted the checkpoint saving and loading procedures, enabling local inference on the available hardware. Our team received a national-level award in the competition. Beyond this project, I have documented my research projects in public repositories, describing the computational workflows and implementation choices to help others understand and use the released materials.

ConvLSTM Prediction of Battery Material Fields

Continuing under Professor Ying Zhao's supervision, I studied how conditional convolutional long short-term memory (ConvLSTM) models could predict the evolution of lithium concentration and von Mises stress fields in battery materials. I automated COMSOL simulations using MATLAB and organized the results into samples containing 25 time-aligned frames of each field, a grain-orientation map, and a C-rate map. I trained separate models in PyTorch for concentration and stress prediction, using a combined mean squared error (MSE) and structural similarity (SSIM) loss. At each prediction step, each model took five consecutive images of its target field together with the grain-orientation and C-rate maps as inputs to predict the next frame. For multi-step prediction, I shifted the five-frame input window forward by one step, appending the predicted frame and removing the oldest frame, while keeping the condition maps unchanged. As predicted frames progressively replaced the reference frames, errors accumulated and eventually produced visible distortion. Drawing on the literature, I introduced scheduled sampling to gradually incorporate model-generated frames into the training windows. I evaluated the resulting concentration and stress sequences against COMSOL reference sequences, examining both image errors and their accumulation over

successive prediction steps. For the sequences examined, training with scheduled sampling reduced image errors over the same prediction horizon and delayed visible distortion compared with training without it.

PINNs for Shock Capturing

Under Professor Xianyang (Tom) Chen's supervision at Tongji University, I studied physics-informed neural networks (PINNs) for shock capturing in the one-dimensional inviscid Burgers equation. I implemented a standard PINN in PyTorch that took spatial and temporal coordinates as inputs to predict the solution $u(x,t)$, using automatic differentiation to compute the derivatives required for the partial differential equation (PDE) residual. The training objective combined PDE-residual, initial-condition, and boundary-condition losses, without using precomputed solution labels. To investigate whether artificial viscosity could improve the representation of the sharp shock transition, I trained a second PINN with a fixed-coefficient artificial-viscosity term added to the residual. I evaluated both models against the analytical inviscid solution, comparing relative L^2 error, maximum pointwise error, shock location, and transition width. The artificial-viscosity model achieved a lower relative L^2 error but a higher maximum pointwise error than the standard PINN. I examined the predicted shock profiles alongside these metrics to assess overall agreement and local errors near the discontinuity. Because the viscosity term changed the governing equation, I interpreted these results as a comparison of agreement with the inviscid reference, considering the overall error and local shock profile together.

Hybrid Finite Element–Neural Operator Solvers

At Johns Hopkins University, I am developing a hybrid finite element–neural operator (FE–NO) solver for solid mechanics under Professor Somdatta Goswami's supervision. The framework uses non-overlapping domain decomposition to divide the computational domain into finite element and neural-operator subdomains connected through a shared interface.

My collaborator and I initially explored predicting nodal displacements and recovering stresses through finite element post-processing. We interpolated the predicted displacements using finite element shape functions, computed displacement gradients using the spatial derivatives of these functions, and obtained strains and stresses through the strain–displacement and material constitutive relations. This stress-recovery procedure amplified prediction errors in our tests, so we chose to predict subdomain stresses directly. I generate paired training data by sampling interface displacement fields using Gaussian random fields (GRFs) and computing the corresponding subdomain stress fields with GPU-accelerated JAX-FEM simulations. Using these data, I train separate Transolver models in JAX, one for each stress component, with interface displacement fields as inputs and the corresponding subdomain stress component as the output. During coupling, the models predict subdomain stresses from the current interface displacements. These stresses are used to compute interface reaction forces for the finite element solver, which updates the interface displacements and passes them back to the neural-operator models for the next iteration. I evaluate the coupled solver by examining stress-prediction accuracy, interface displacement compatibility, force balance, and convergence. When assessing computational efficiency, I distinguish the cost of a single neural-operator evaluation from the total cost of obtaining a coupled solution at the required accuracy.

After training, the Transolver models predicted stresses accurately on held-out interface displacement fields sampled from GRFs, but their prediction errors increased substantially when used with interface displacements generated during FE–NO coupling. To investigate this discrepancy, I compared the displacement fields encountered during coupling with those sampled for training and found that the sampled data did not adequately represent the actual coupling inputs. Using the coupling-generated displacement fields as a reference, I adjusted the GRF length scale and amplitude, generated new interface displacement samples, and computed their corresponding subdomain stress fields with JAX-FEM. I then retrained the models on the revised paired dataset. After retraining, the models retained good stress-prediction accuracy on the GRF validation set and showed lower errors on coupling-generated inputs in cylinder and bracket tests. I used these coupled tests to assess the revised data-generation strategy alongside validation on sampled data. I also investigated convergence acceleration and interface relaxation strategies for the iterative coupling process.

Building on this work, I am extending FE–NO coupling to a three-dimensional 100-fiber composite with non-overlapping fiber and matrix subdomains. FEM solves the matrix, while separate Transolver models predict each stress component in the fibers, with component-specific weights shared across all fibers. The subdomains iteratively exchange interface displacements and reaction forces. We apply periodic boundary conditions along the fiber direction to eliminate fiber-end effects.

Future Work

My future research interests center on combining neural operators and numerical solvers for mechanics. Building on my experience with FE–NO coupling, I would like to study how interface coupling strategies and neural-operator approximation errors affect the accuracy and convergence of the coupled solution. I am also interested in whether an operator trained at one mesh resolution can be used on finer or coarser meshes, and how changes in resolution affect prediction accuracy and computational cost. My experience with JAX and JAX-FEM has also drawn me to GPU-accelerated scientific computing and differentiable simulation. I would like to connect numerical solvers and surrogate models within a computational graph, using automatic differentiation and gradient backpropagation to compute how physical objectives depend on material and geometric parameters for structural optimization and inverse problems. In particular, I hope to investigate how gradients propagate through iterative coupling and how solver convergence and surrogate-model errors affect gradient accuracy. To pursue these directions, I seek a stronger foundation in mechanics, numerical analysis, and optimization, together with training in GPU parallel computing to implement these methods efficiently for larger three-dimensional problems.

My long-term goal is to pursue research and development in scientific computing, including opportunities in industrial research teams such as those at NVIDIA. I would like to work on GPU-accelerated numerical methods and differentiable simulation, particularly for structural optimization and inverse problems, and participate in the development of scientific computing software. Through doctoral training, I hope to build the theoretical foundation and research experience needed to develop these methods, evaluate their accuracy and computational cost, and implement them for engineering applications.